HW1 Report: Kinematics and Dynamics of Rotation

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1 Task 1: Lie Integration of Quaternion Kinematics

1.1 Problem Statement

A rotating rigid body is parameterized by a quaternion

$$q = [q_0, q_1, q_2, q_3]^T$$

and follows the kinematic equation

$$\dot{q} = \frac{1}{2} q \otimes \hat{\omega}, \quad \hat{\omega} = [0, \omega_x, \omega_y, \omega_z]^T. \tag{1}$$

The exact discrete update used in the simulation is

$$q_{k+1} = q_k \otimes \exp\left(\frac{1}{2}\hat{\omega}_k \, dt\right). \tag{2}$$

For comparison, the Forward Euler update is

$$q_{k+1}^{\text{Euler}} = q_k + dt \, \dot{q}_k. \tag{3}$$

1.2 Numerical Setup

- Final time: $T = 12\,\text{s}$
- Time step: $dt = 0.02\,\text{s}$
- Angular velocity: $\omega = [0.7,\ 1.1,\ 0.5]^T\,\text{rad/s}$
- Initial quaternion: $q_0 = [1, 0, 0, 0]^T$

1.3 Results

The Lie integrator preserves the unit norm better than Forward Euler.

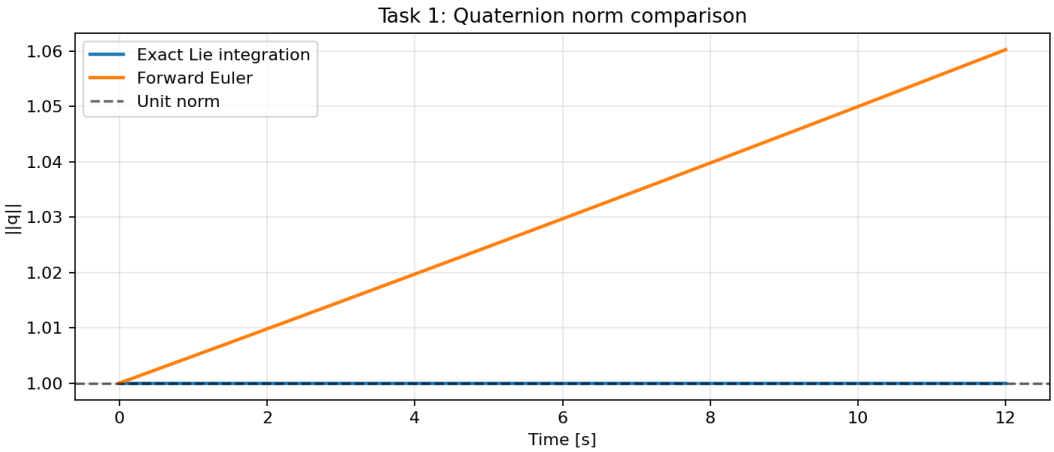

Figure 1: Task 1: Quaternion norm comparison (Lie exact update vs Forward Euler).

Quaternion component trajectories are shown below.

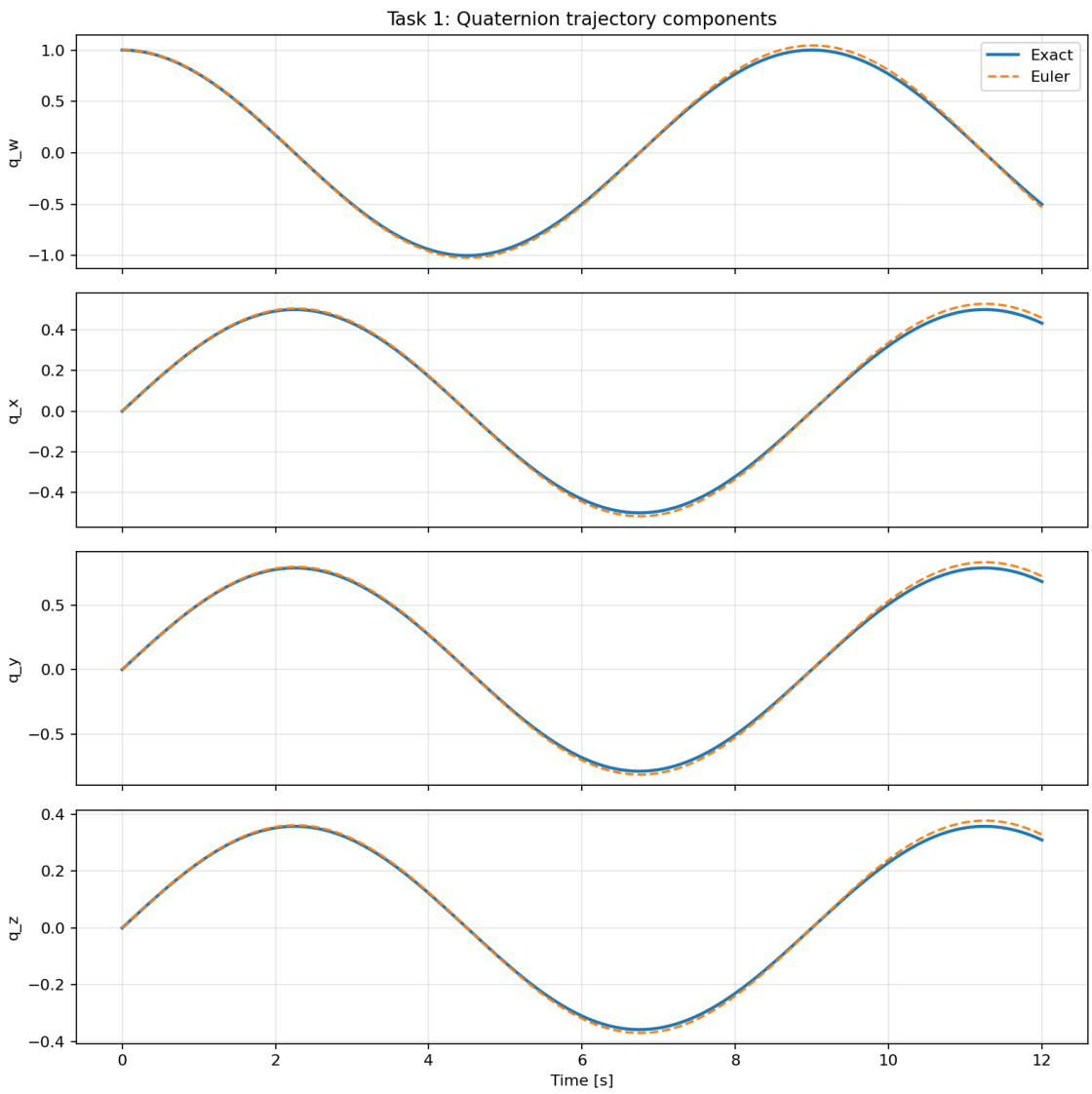

Figure 2: Task 1: Quaternion component trajectories.

Exported animations:

- `outputs/rotation_exact.mp4`
- `outputs/rotation_euler.mp4`

1.4 Conclusion

- Exact Lie integration keeps $\|q\|$ close to 1.
- Forward Euler introduces norm drift and orientation error over time.

2 Task 2: Newton-Euler Dynamics Integration

2.1 Problem Statement

Simulate rigid-body translational and rotational dynamics using Lie integration for quaternion kinematics, and provide plots of:

- position (x, y, z) and linear velocity,
- orientation (quaternion components) and angular velocity.

2.2 Dynamic Model

State vector:

$$x = [q,\, \omega,\, p,\, v]^T. \tag{4}$$

Governing equations:

$$\dot{p} = v, \tag{5}$$

$$\dot{v} = g + \frac{1}{m} R(q) F_{\text{body}}, \tag{6}$$

$$\dot{\omega} = J^{-1}(\tau - \omega \times (J\omega))\,. \tag{7}$$

Quaternion update on the Lie group:

$$q_{k+1} = q_k \otimes \exp\left(\frac{1}{2}\hat{\omega}_{\text{mid}}\, dt\right), \quad \omega_{\text{mid}} = \frac{\omega_k + \omega_{k+1}}{2}. \tag{8}$$

2.3 Numerical Setup

- Final time: $T = 12\,\text{s}$
- Time step: $dt = 0.01\,\text{s}$
- Mass: $m = 1.8\,\text{kg}$
- Inertia: $J = \text{diag}(2.0,\ 1.3,\ 0.8)$
- Gravity: $g = [0, 0, -9.81]^T\ \text{m/s}^2$
- Body force: $F_{\text{body}} = [4, 0, 18]^T$
- Body torque: $\tau = [0.15, -0.05, 0.10]^T$
- Initial conditions:
  - $q_0 = [1, 0, 0, 0]^T$
  - $\omega_0 = [0.6, 1.0, 0.3]^T$
  - $p_0 = [0, 0, 0]^T$
  - $v_0 = [1.0, 0.4, 0.2]^T$

2.4 Results

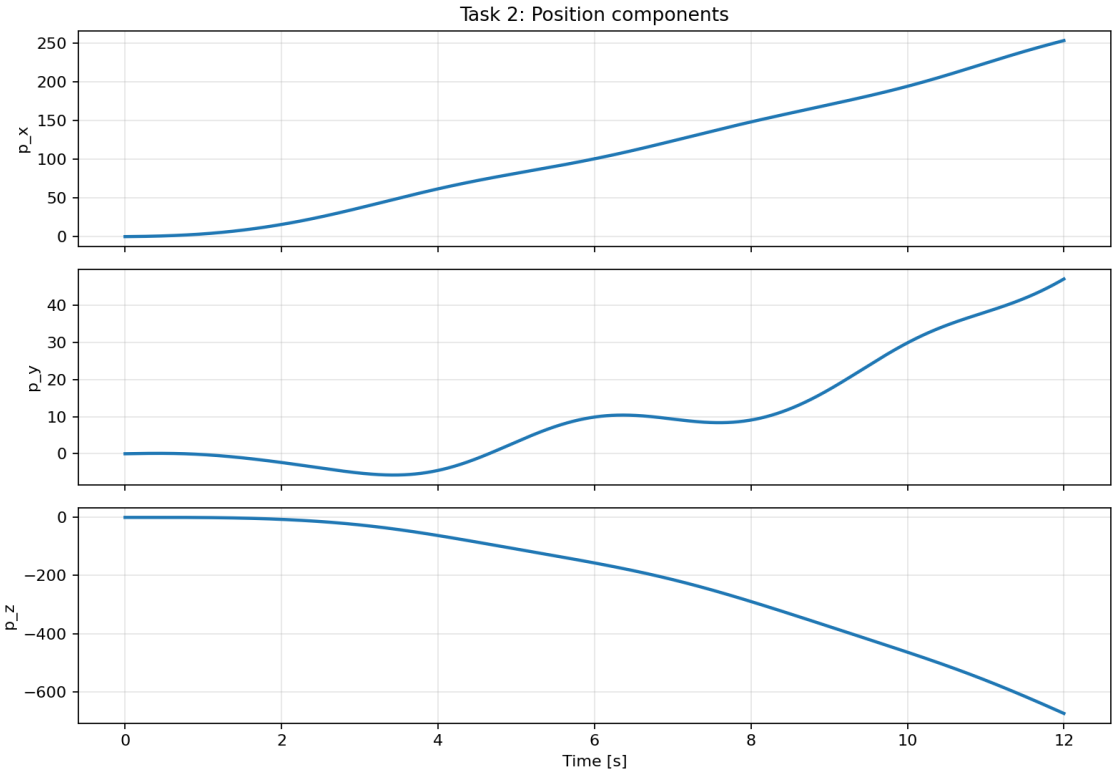

Figure 3: Task 2: Position components.

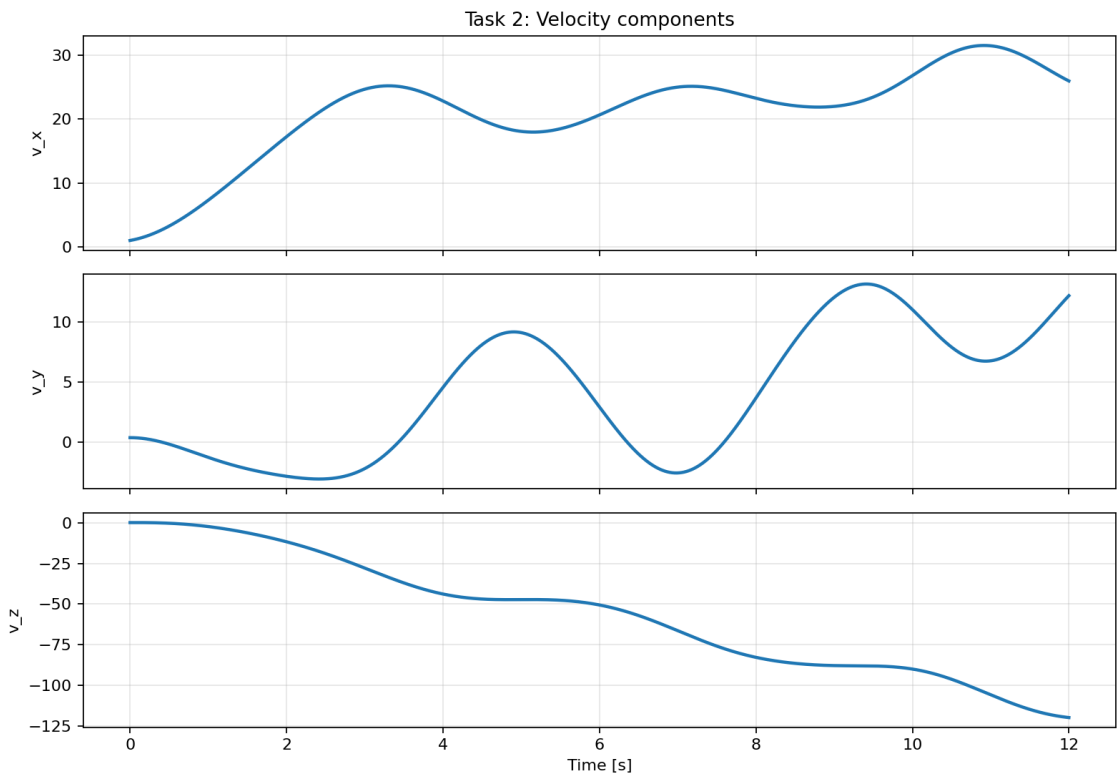

Figure 4: Task 2: Velocity components.

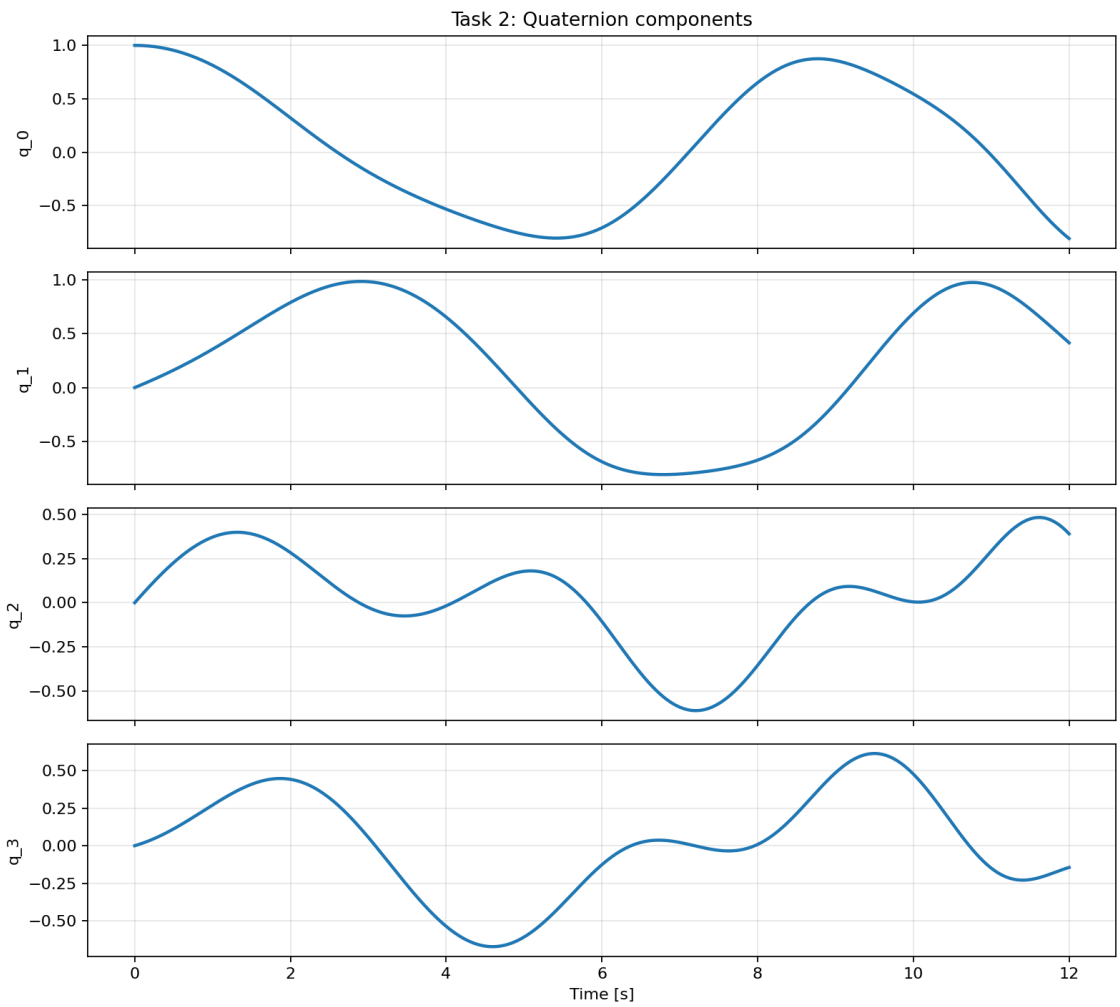

Figure 5: Task 2: Quaternion components.

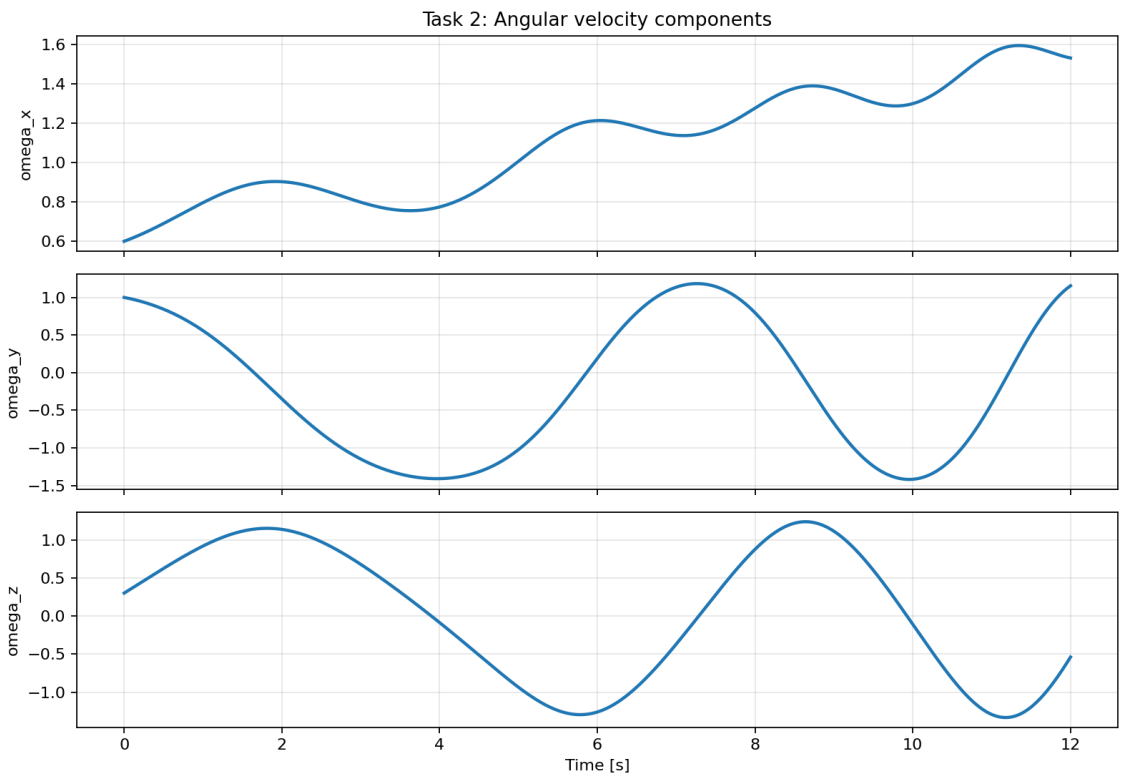

Figure 6: Task 2: Angular velocity components.

Exported animation:

- `outputs/rigid_body_motion_problem2.mp4`

2.5 Conclusion

- The simulation shows coupled translational and rotational motion under constant body-frame force and torque.
- Quaternion norm remains near unity due to Lie-group integration.